

Отчёт по лабораторной работе №3

Управляющие структуры

Владимир Базлов

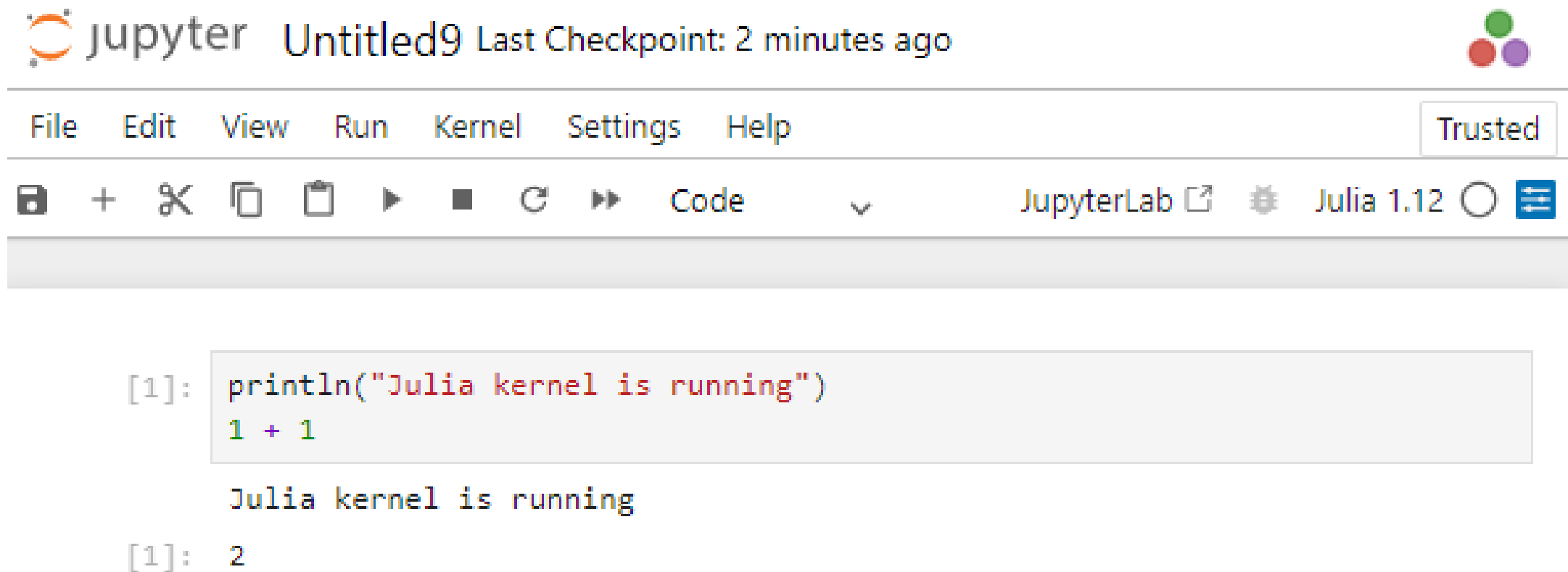
Освоить применение циклов, функций и сторонних пакетов Julia для решения задач линейной алгебры и работы с матрицами.

Задание 8 — реализовать функцию `outer(x, y, operation)` и построить матрицы заданной структуры.

Задание 9 — решить систему линейных уравнений и обобщить решение.

Задание 10 — создать и проанализировать матрицу $M_{6 \times 10}$.

Задание 11 — вычислить двойные суммы двумя способами.



The image shows a JupyterLab interface. At the top, there's a header bar with the Jupyter logo, the text "jupyter", and "Untitled9 Last Checkpoint: 2 minutes ago". To the right of the header is a "Trusted" button. Below the header is a menu bar with "File", "Edit", "View", "Run", "Kernel", "Settings", and "Help". Below the menu bar is a toolbar with icons for saving, adding, deleting, copying, pasting, running, and other actions. To the right of the toolbar are links for "JupyterLab", a settings gear, "Julia 1.12", and a help icon. The main area contains a code cell with the following content:

```
[1]: println("Julia kernel is running")
      1 + 1

      Julia kernel is running
```

Below the code cell, the output is displayed:

```
[1]: 2
```

```
[3]: n = 0  
while n < 10  
    n += 1  
    println(n)  
end
```

```
1  
2  
3  
4  
5  
6  
7  
8  
9  
10
```

```
[5]: myfriends = ["Ted", "Robyn", "Barney", "Lily", "Marshall"]

i = 1
while i <= length(myfriends)
    friend = myfriends[i]
    println("Hi $friend, it's great to see you!")
    i += 1
end
```

```
Hi Ted, it's great to see you!
Hi Robyn, it's great to see you!
Hi Barney, it's great to see you!
Hi Lily, it's great to see you!
Hi Marshall, it's great to see you!
```

```
[7]: for n in 1:2:10  
      println(n)  
      end
```

```
1  
3  
5  
7  
9
```

```
[9]: myfriends = ["Ted", "Robyn", "Barney", "Lily", "Marshall"]  
  
for friend in myfriends  
    println("Hi $friend, it's great to see you!")  
end
```

```
Hi Ted, it's great to see you!  
Hi Robyn, it's great to see you!  
Hi Barney, it's great to see you!  
Hi Lily, it's great to see you!  
Hi Marshall, it's great to see you!
```

```
[11]: x = 5  
      y = 10  
      (x > y) ? x : y
```

```
[11]: 10
```

```
[13]: function f1(x)
      x^2
      end

      f2(x) = x^2

      f3 = x -> x^2

      f1(3), f2(3), f3(3)
```

```
[13]: (9, 9, 9)
```

```
[15]: v = [3, 5, 2]
      sort(v)
      println(v)

      sort!(v)
      println(v)

      [3, 5, 2]
      [2, 3, 5]
```

```
[17]: map(x -> x^3, [1, 2, 3])
```

```
f(x) = x^2
```

```
f.([1, 2, 3])
```

```
[17]: 3-element Vector{Int64}:
```

```
1
```

```
4
```

```
9
```

```
[1]: import Pkg
      Pkg.add("Colors")
      using Colors

      palette = distinguishable_colors(100)
      rand(palette, 3, 3)
```

```
Warning: attempting to remove probably stale pidfile
path = "C:\\Users\\Asus\\.julia\\registries\\.pid"
@ FileWatching.Pidfile C:\\Users\\Asus\\AppData\\Local\\Programs\\Julia-1.1
2.0\\share\\julia\\stdlib\\v1.12\\FileWatching\\src\\pidfile.jl:300
Resolving package versions...
Project No packages added to or removed from `C:\\Users\\Asus\\.julia
\\environments\\v1.12\\Project.toml`
Manifest No packages added to or removed from `C:\\Users\\Asus\\.julia
\\environments\\v1.12\\Manifest.toml`
```



```
[3]: function outer(x, y, operation)
      m = size(x, 1)
      k = size(x, 2)
      n = size(y, 2)

      C = zeros(eltype(x), m, n)
      for i in 1:m
          for j in 1:n
              s = zero(eltype(x))
              for t in 1:k
                  s += operation(x[i, t], y[t, j])
              end
              C[i, j] = s
          end
      end
      return C
  end
```

```
[3]: outer (generic function with 1 method)
```

```
[5]: A = [1 2 3; 4 5 6]  
     B = [1 0; 0 1; 1 1]
```

```
C1 = outer(A, B, *)
```

```
C2 = A * B
```

```
println(C1)
```

```
println(C2)
```

```
println(C1 == C2)
```

```
[4 5; 10 11]
```

```
[4 5; 10 11]
```

```
true
```

```
[7]: function structure_matrix(n)
      M = zeros(Int, n, n)
      for i in 1:n
          M[i, i] = 2
          i > 1 && (M[i, i-1] = -1)
          i < n && (M[i, i+1] = -1)
      end
      return M
    end

    structure_matrix(5)
```

```
[7]: 5x5 Matrix{Int64}:
      2  -1   0   0   0
     -1   2  -1   0   0
      0  -1   2  -1   0
      0   0  -1   2  -1
      0   0   0  -1   2
```

```
[9]: structure_matrix(10)
```

```
[9]: 10x10 Matrix{Int64}:
```

2	-1	0	0	0	0	0	0	0	0
-1	2	-1	0	0	0	0	0	0	0
0	-1	2	-1	0	0	0	0	0	0
0	0	-1	2	-1	0	0	0	0	0
0	0	0	-1	2	-1	0	0	0	0
0	0	0	0	-1	2	-1	0	0	0
0	0	0	0	0	-1	2	-1	0	0
0	0	0	0	0	0	-1	2	-1	0
0	0	0	0	0	0	0	-1	2	-1
0	0	0	0	0	0	0	0	-1	2

```
[11]: using LinearAlgebra

n = 5
A = zeros(Float64, n, n)
for i in 1:n
    A[i, i] = 2.0
    i > 1 && (A[i, i-1] = -1.0)
    i < n && (A[i, i+1] = -1.0)
end

y = ones(Float64, n)
x = A \ y

println("A = "); display(A)
println("y = ", y)
println("x = ", x)
println("A*x = ", A * x)

A =
5x5 Matrix{Float64}:
 2.0 -1.0  0.0  0.0  0.0
-1.0  2.0 -1.0  0.0  0.0
 0.0 -1.0  2.0 -1.0  0.0
 0.0  0.0 -1.0  2.0 -1.0
 0.0  0.0  0.0 -1.0  2.0
y = [1.0, 1.0, 1.0, 1.0, 1.0]
x = [2.5, 3.9999999999999996, 4.499999999999999, 4.0, 2.5]
A*x = [1.0000000000000004, 1.0, 0.9999999999999982, 1.0000000000000009,
1.0]
```

```
[13]: function solve_system(n)
      A = zeros(Float64, n, n)
      for i in 1:n
          A[i, i] = 2.0
          i > 1 && (A[i, i-1] = -1.0)
          i < n && (A[i, i+1] = -1.0)
      end
      y = ones(Float64, n)
      return A \ y
    end

    solve_system(10)
```

```
[13]: 10-element Vector{Float64}:
      4.999999999999999
      8.999999999999998
     11.999999999999998
     14.0
     15.0
    15.000000000000002
     14.0
     12.0
      9.0
      5.0
```

```
[15]: using Random
      Random.seed!(42)

      M = rand(1:10, 6, 10)
      display(M)
```

6x10 Matrix{Int64}:

7	7	4	5	4	4	7	9	8	4
5	7	3	5	4	2	5	2	7	2
5	5	6	1	5	8	8	3	3	1
8	3	1	6	9	4	2	10	4	9
7	7	3	2	4	1	6	4	6	10
2	7	2	7	6	1	5	8	7	3

[]:

|



```
[17]: N = 4  
counts = [count(>(N), M[i, :]) for i in 1:size(M, 1)]  
println("Число элементов > $N по строкам: ", counts)
```

```
Число элементов > 4 по строкам: [6, 5, 6, 5, 5, 6]
```

```
[19]: M_val = 7  
rows_two = [i for i in 1:size(M, 1) if count(==(M_val), M[i, :]) == 2]  
println("Строки, где $M_val встречается ровно 2 раза: ", rows_two)
```

Строки, где 7 встречается ровно 2 раза: [2, 5]

```
[21]: K = 75
ncols = size(M, 2)
pairs = Tuple{Int,Int}[]
for j1 in 1:ncols
    for j2 in (j1+1):ncols
        if sum(M[:, j1]) + sum(M[:, j2]) > K
            push!(pairs, (j1, j2))
        end
    end
end
println("Пары столбцов с суммой > $K: ", pairs)
```

```
Пары столбцов с суммой > 75: Tuple{Int64, Int64}[]
```

```
[23]: S1 = 0.0
      for i in 1:20
        for j in 1:5
          S1 += i^4 / (3 + j)
        end
      end
      println("S1 = ", S1)
```

```
S1 = 639215.2833333334
```

```
[25]: S2 = 0.0
      for i in 1:20
        for j in 1:5
          S2 += i^4 / (3 + i*j)
        end
      end
      println("S2 = ", S2)
```

```
S2 = 89912.02146097136
```

```
[27]: S1_alt = sum(i^4 / (3 + j)    for i in 1:20, j in 1:5)
      S2_alt = sum(i^4 / (3 + i*j) for i in 1:20, j in 1:5)
      println(S1_alt, " ", S2_alt)
```

```
639215.2833333338  89912.02146097131
```

1. Изучены управляющие структуры Julia: циклы `while` и `for`, тернарный оператор и условные выражения.
2. Освоены три способа объявления функций, включая анонимные функции и функции высшего порядка.
3. Изучены `map` и `broadcast`, а также работа со сторонним пакетом `Colors`.
4. Реализована собственная функция `outer` и показана её эквивалентность матричному умножению при операции `*`.
5. Освоено решение систем линейных уравнений матричным методом и обобщение решения на произвольную размерность.
6. Освоен анализ матриц и вычисление двойных сумм циклами и генераторами.
7. Проверена воспроизводимость результатов в JupyterLab.